



Note: i. Answer **all** questions. ii. Assume suitable data if necessary iii. Use of non-programmable calculators is permitted. iv. This paper contains one page only (back to back).

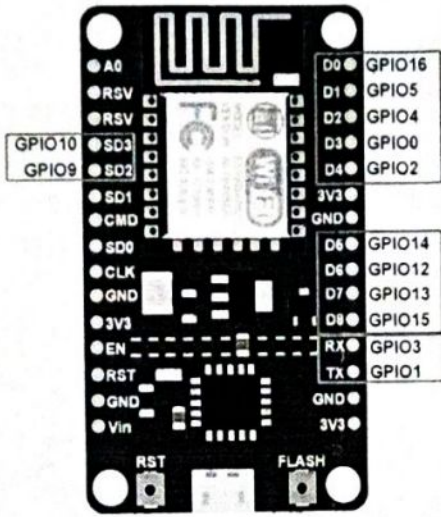
Q.N.	Question	Marks	COs	BT
Q.1	c. Write a short note on Architecture of IoT. d. Differentiate between IOT and Industrial IoT.	10	1, 2	BT 1, 2,
Q.2	Name the device shown in figure (i). Redraw the figure (i) and label its parts. List its features. 	10	2	BT 1, 5, 6

Figure: (i)

*****!!!!!!!!!!!!All the very Best!!!!!!!!!!!!*****

Details below are for Information only

Bloom's Taxonomy

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create
COs: 1. To understand the architectural overview of IoT, 2. To understand the various sensors, actuators, and embedded platforms, 3. To understand various Internet protocols for IoT, 4. To understand different cloud platform services, 5. To understand real world IoT Applications and design constraints

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	ISE-II Examination (Semester-I/ VII) 2025-26	
	Final Year B. Tech.	Electronics and Telecommunication Engineering
	Entrepreneurial Economics - EE	HMC - EC409
Date: 14/Oct/2023 (FRI)	Time: 14:30 to 15:15	Max. Marks : 20

Note: i. Assume suitable data if necessary. ii. Use of non-programmable calculators is permitted.
iii. This paper contains HALF page only. iv. Solve all questions.

Q.N	Question	Marks	C O	BT
Q.1	List at least five intellectual property types with laws governing the same and life of it.	05	4, 5	1, 2
Q.2	Define Innovation. What is Patentable subject matter under the Act with at least one example? Name the related Act.	05	1, 4	1, 2
Q.3	How does economics play the role of a tool for entrepreneurs to make better decisions?	05	3	2
Q.4	Write 'Gains of Trade' concept with example of labor division specialization? 3	05	2	4

*****!!!!!!All the Best!!!!!!*****

Details below are for Information only: **Bloom's Taxonomy:** BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

Course outcomes: 1. Define entrepreneurship 2. Identify examples of entrepreneurship 3. Articulate the traits of an entrepreneur 4. Describe the importance of the entrepreneur's role in society 5. Provide examples of how to serve needs in their own community through entrepreneurial action

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electronics and Telecommunication In-Semester Examination II Academic Year: 2025-26 (Semester-I)		
Class: Final Year B.Tech.	Course: Data Communication and Networking	Course Code: PEC-EC-403	
Date: 10.10.2025	Time: 14:30 to 15:15	Max. Marks: 20	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Find the CRC for 1110010101 with the divisor x^3+x^2+1 .	05	CO3	BT3
Q.2	Host A wants to send 10 frames to Host B. The Hosts agreed to go with Go-Back-4. How many number of frames are transmitted by Host A if every 6 th frame that is transmitted by Host A is either corrupted or lost?	05	CO3	BT3
Q.3	What is Wireless LAN (WLAN)? Explain its Architecture and Advantages.	05	CO2	BT1
Q.4	Compare X.25, Frame Relay and ATM Technologies.	05	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the components and infrastructure that form the basis for most computer networks
CO 2	Describe the technical aspects of data communications on the Internet
CO 3	Achieve practical experience in network programming.
CO 4	Propose network designs based on case studies in colleges or other institutions
CO 5	Recognize security issues in a network, Design and Evaluate short path algorithm.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: Final Year B.Tech. Course Name: Electronic Instrumentation Course Code: PCC-EC402
Date: 11.10.2025 Time: 14:30 to 15:15 Max. Marks: 20

Note:

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- The symbols carry usual meaning.

Que. No.	Question	Marks	CO	BT
Q.1	Derive equation for gauge factor of wire strain gauges.	5	CO2	BT4
Q.2	A linear resistance potentiometer is 50 mm long and is uniformly wound with a wire having resistance of 10 K Ω . Under normal condition, the slider is at the centre of the potentiometer. Find the linear displacement when the resistance of the potentiometer as measured by Wheatstone bridge for two cases is (i) 3850 Ω , (ii) 7560 Ω . Are the two displacements in the same direction?	5	CO3	BT3
Q.3	State and explain the laws of thermoelectricity.	5	CO2	BT2
Q.4	Prove that the differential arrangement of parallel plate capacitive transducer exhibits linear characteristics.	5	CO2	BT3

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication
Academic Year: 2025-26 (semester I)
In-Semester Examination 2

Class: Final Year B.Tech.

Course Name: Microwave Engineering

Course Code: PCC-EC-401

Date: 09.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

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- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Consider an air – filled rectangular waveguide with dimensions $a = 2.286$ cm and $b = 1.016$ cm. The increasing order of the cut – off frequency for different modes is (a) $TE_{01} < TE_{10} < TE_{11} < TE_{20}$ (b) $TE_{20} < TE_{11} < TE_{10} < TE_{01}$ (c) $TE_{10} < TE_{20} < TE_{01} < TE_{11}$ (d) $TE_{10} < TE_{11} < TE_{20} < TE_{01}$	5	CO1	BT5
	b	State Poynting theorem and derive an expression for power density and average power density of EM wave.	5	CO2	BT1
Q.2	a	The time domain expression for the magnetic field at plane wave travelling in a nonmagnetic medium is given by $H(x, t) = 0.7 \cos(6\pi \times 10^8 t - 10x) \hat{a}_z$ A/m. Find i) direction of wave of propagation, ii) dielectric constant and intrinsic impedance iii) time domain expression of E- field iv) average power density	5	CO2	BT5
	b	Compare rectangular waveguide and circular waveguide.	5	CO1	BT4
Q.3	a	A rectangular waveguide has a cross sectional dimension at $a = 6$ cm and $b = 3$ cm. determine the modes which will propagate through the waveguide at a frequency of 5 GHz	5	CO3	BT5
	b	Calculate λ_c , λ_g , f_c , v_p , v_g and impedance of a given circular waveguide whose internal diameter is 5.5 cm used for a signal at a frequency of 12 GHz propagated in TE_{01} mode. (Data: Bessel's equation value for TE_{01} mode = 3.832)	5	CO3	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the knowledge of waveguide behavior to solve practical problems in microwave communication systems and waveguide-based devices.
CO 2	Analyze electromagnetic systems using Poynting's theorem to determine the distribution of energy and evaluate power transfer in waveguides, transmission lines, and antennas.
CO 3	Calculate the cut-off frequencies, phase velocity, group velocity, and wavelength for different modes in rectangular waveguides